St. Philomena's College (Autonomous), Mysore
PG Department of Studies and Research in Chemistry
Question Bank (Revised Curriculum 2018 onwards)
First Year- Second Semester (2019 -21 Batch)
Coordination Chemistry QP Code: 55102

Unit	SI. No	Questions	Marks
1	1.	Give an example each for complexes with coordination number 5 and 7. Write their structure.	2
1	2.	How are complex formation detected? Explain two methods with examples.	2
1	3.	How are complex formation detected? Explain any two methods.	2
1	4.	Predict the coordination number and geometries for the following complexes $[V(acac)_2]$ and $[Cu(NH_3)_4]^{2+}$	2
1	5.	What is the geometry possible for coordination number six? Give examples.	2
1	6.	Sketch the two structures that describe most six-coordinate complexes. Which is more common ? Explain with suitable examples.	2
1	7.	What is the geometry possible for coordination number four? Give examples.	2
1	8.	Give an example of a pentanuclear cluster and write the structures	2
1	9.	How are stepwise and overall stability constants related?	2
1	10.	Distinguish between kinetic and thermodynamic stability of a metal complex.	2
1	11.	$[Fe(H_2O)_6]^{3+}$ and $[Cr(H_2O)_6]^{3+}$ are equally stable thermodynamically but former is labile whereas the latter is inert. Justify.	2
1	12.	As a general trend the step-wise stability constants gradually decrease. Comment.	2
1	13.	Define stepwise stability constant and overall stability constant.	2
1	14.	why does stablity constant value of a complex gradually decreases?	2
1	15.	Write the factors affecting the stability of complexes.	2
1	16.	Explain Irwing –William series.	2
1	17.	Why $[Co(en)_3]^{3+}$ is more stable than $[Co(NH_3)_6]^{3+}$? .	2
1	18.	Define chelate effect.	2

1	19.	List the factors affecting the stability of the complexes.	2
1	20.	What do you understand by the stability of the chelate ring?	2
1	21.	How does the nature of ligand affect the stability constant?	2
1	22.	How does the nature of the central metal ion effect the stability of the complex?	2
1	23.	How chelation increases the stability of the complex?	2
1	24.	Explain the chelate effect on the basis of entropy change?	2
1	25.	What is macrocyclic effect? Explain.	2
1	26.	In the series of complexes of iron(II) with phenanthroline, K3 > K2. Why?	2
1	27.	Mention the methods adopted to determine the stability constants of the complexes?	2
1	28.	Rationalize the fact that stability increass in the order; [Co(NH3)6]3+ < [Co(en3)]3+ < [Co(dien)2]3+	2
1	29.	[Cu(en)2 (H2O)2]2+, even though it contains only two chelate rings , is more stable than [Cu(en)3]2+ which contains three chelate rings. Comment.	2
1	30.	log K1 for Ni2+ - en complex is 7.9, whereas that for Ni2+ - trien is 14.0. Comment.	2
1	31.	How does the size of the chelate rings affect the stability of the complexes?	2
1	32.	How does the number of the chelate rings affect the stability of the complexes?	2
1	33.	Give two examples for the preparative methods of coordination compounds.	2
1	34.	Mention two examples from redox reaction of preparation of coordination compounds.	2
1	35.	Give an example for the preparation of coordination compunds by thermal dissociation reaction.	2
2	36.	What are high spin and low spin complexes? Explain with suitable examples.	2
2	37.	The hexaaquamanganese(II) ion contains five unpaired electrons, while the hexacyano ion contains only one unpaired electron. Explain using CFT.	2

2	38.	For $[Cr(H_2O)_6]^{2+}$ ion EP = 23500 cm-1 and Δo = 13900 cm-1. Will it be a high spin or low spin complex?	2
2	39.	Arrange d orbitals splitting in the increasing order of their energies for the square planar and tetragonal complexes.	2
2	40.	Among the complex ion, $[Fe(H_2O)_6]^{2+}$ and $[Fe(H_2O)_6]^{3+}$, which is expected to give greater splitting of d orbitals? Explain.	2
2	41.	What is CFSE? Calculate the CFSE for $[Co(H_2O)_6]^{3+}$and $[Co(NH_3)_6]^{2+}$.	2
2	42.	Write the increasing order of d orbital splitting in square planar complexes.	2
2	43.	Differentiate between high spin and low spin complexes with an example.	2
2	44.	For $[Cr(H_2O)_6]^{2+}$ ion, EP= 23500 cm-1 and Δo = 13900 cm-1. Will it be high or low spin.	2
2	45.	What are the postulates of crystal field theory?	2
2	46.	What are the limitations of crystal field theory?	2
2	47.	What is CFSE? Calculate the CFSE for d3 ion.	2
2	48.	Tetrahedral spins are always high spin, why?	2
2	49.	CFSE of d^{10} system is zero. Why?	2
2	50.	Calculate the CFSE for d4, d6 HS and LS octahderal complex.	2
2	51.	Calculate the CFSE for d9 tetrahedral and octahdral complex.	2
2	52.	Mention the factors affecting 10Dq value.	2
2	53.	What is a Jahn Teller effect? Explain with example.	2
2	54.	What is spectrochemical series? Why is it so called?	2
2	55.	Which ion is expected to have Jahn-Teller distortion in octahedral field? Co^{3+}(strong field); Co^{2+} (low spin); Mn^{2+}; Ni^{2+}(Weak field).	2
2	56.	Account for Δo increase in the order for the following complexes. $[Co(H_2O)_6]^{2+} < [Co(H_2O)_6]^{3+} < [Rh(H_2O)_6]^{3+} < [Ir(H_2O)_6]^{3+}$.	2
2	57.	Which of the complex has higher Δo value and why? a) $[CoCl_6]^{3-}$ b) $[Cr(CN)_6]^{3-}$.	2

2	58.	For the same ligand, the value of Δo in octahedral complexes of Co^{3+}, Rh^{3+},Ir^{3+} are approximately in the ration of 2 : 3 : 4, Give reason.	2
2	59.	Determine the number of unpaired electrons in $[Cr(NH_3)_6]^{3+}$ ion.	2
2	60.	Give one evidence for metal ligand orbital overlap.	2
2	61.	Define CFSE. Calculate CFSE for d7 , low spin and high spin octahedral system.	2
2	62.	What is spectrochemical series? Mention any two applications.	2
2	63.	With the help of MOT calculate the spin only magnetic moment of Co^{3+} octahedral complex with strong and weak ligand field.	2
2	64.	With the help of MOT calculate the spin only magnetic moment of Co^{2+} in tetrahedral field.	2
2	65.	What is term symbol? Give the ground term for $3d^6$ of Fe^{2+}.	2
2	66.	Illustrate that d^n and d^{10-n} configurations have same term symbols.	2
2	67.	What is term symbol? Give the ground term for $3d^6$ of Fe^{2+}.	2
2	68.	What are the important selection rules for the electronic spectra of metal complexes?	2
2	69.	Find the ground term for free ion with d^8 and d^9 configurations.	2
2	70.	What are spin multiplicity forbidden and Laporte forbidden transitions?	2
2	71.	What are selection rules for electronic spectra?	2
2	72.	What is the ground state term of d^2 configuration?	2
2	73.	Calculate the Russel- Saunders term for p^2 configuration?	2
2	74.	Calculate the Russel- Saunders term for d^6 configuration?	2
2	75.	Draw the electronic spectra of $[Ni(H_2O)_6]^{2+}$.	2
2	76.	Identify the ground state terms from the following $^3F_2$, $^3F_3$, $^3F_4$	2
2	77.	Identify the ground state terms from the following ^1S, ^2P, ^2D, ^3F, ^1G	2
2	78.	Identify the ground state terms from the following ^1S, ^3P, ^1D, ^3F, ^1G	2

2	79.	Commen on the electronic spectrum of $Ti(H_2O)_6]^{3+}$	2
2	80.	Find out the ground state term with spin multiplicity for Cu^{2+} and V^{3+} ions.	2
2	81.	Discuss the special feature of electronic spectra of Mn^{2+} octahedral complex ion.	2
2	82.	State and explain laporte selection rule?	2
2	83.	Why do tetrahedral complexes of d-block elements give much more intense d-d spectra than its O_h complexes	2
2	84.	Draw the orgel diagram for the Cr(III) octahedral complexe.	2
2	85.	What are orgel diagram? Draw the orgel diagram for $Cu(H_2O)_6]^{2+}$	2
2	86.	What are the limitations of Orgel diagrams?	2
2	87.	What are the limitations of Orgel energy level diagrams?	2
2	88.	What are orgel diagrams? What information is conveyed by these diagrams?	2
2	89.	Draw the orgel diagram of d^1 system.	2
2	90.	What are Tanabe Sugano diagram? How does it differ from orgel diagrams.	2
2	91.	Draw the Orgel diagram for d^2 configeration in Oh complexe?	2
2	92.	Draw the Orgel diagram for d^2 configeration in tetrahedral complexe?	2
2	93.	Aqueous solution of $KMnO_4$ is intensely colored. Why?	2
2	94.	What is charge transfer spectra?	2
2	95.	what is the origin of bright yellow colour of $CrO_4^{2-?}$	2
2	96.	Explain why the $[Mn(H_2O)_6]^{2+}$ ion has an extremely pale pink colour.	2
2	97.	Account for the intense colour exhibited by $KMnO_4$ and CrO_4^{2-}.	2
2	98.	What is nephlauxtic ratio? Give its significance.	2
2	99.	What is nephlauxetic effect?	2
2	100.	Why Mn(II), d^5 metal complexes are colourless?	2
2	101.	Which would you expect to have greater magnetic moment, $CoCl_4^{2-}$ or CoI_4^{2-} why?.	2

2	102.	Predict the number of unpaired electrons, the spin only magnetic moments at 25^0C for each of the $[Fe(CN)_6]^{4-}$ and $[Ru(NH_3)_6]^{3+}$.	2
2	103.	$[NiCl_4]^2$? is paramagnetic while $[Ni(CO)_4]$ is diamagnetic though both are tetrahedral. Why?	2
2	104.	Calculate the spin-only μeff values for $K_3[Fe(CN)_6]$ and $[Fe(H_2O)_6]Cl_3$.	2
2	105.	Name the calibrant used in Gouy's method and why it is preferred?	2
2	106.	If Fe3+ form s high-spin octahedral complex, what is its magnetic moment.	2
2	107.	$[Co(CN)_6]^{3-}$ is diamagnetic but $[Co(F)_6]^{3-}$ is paramagnetic with the magnetic moment 4.9 BM. Account for this difference in magnetic properties of these two octahedral complexes using CFT.	2
2	108.	Using CFT, show the orbital occupancies for both weak and strong field octahedral fields for Mn2+, Zn2+, Fe2+ and Co2+. Calculate the spin-only magnetic moment of the ions.	2
2	109.	Using CFT, show the orbital occupancies for both weak and strong field octahedral fields for Mn^{2+}, Zn^{2+}, Fe^{2+} and Co^{2+}. Calculate CFSE values.	2
2	110.	$Co(NH_3)_6]^{3+}$ is yellow and diamagnetic, while $[Co(F)_6]^{3-}$ is blue and paramagnetic why?	2
2	111.	Which one of the following ions has maximum magnetic moment; Cr^{2+}, Fe^{3+}, Ni^{2+}	2
2	112.	what is magnetic susceptibility? Name a method used to measure it.	2
2	113.	Calculate the spin only magnetic moment of Co^{2+} ion V3+ ion?	2
2	114.	Which ion exhibit a larger magnetic moment: Mn^{2+} or V^{2+}?	2
2	115.	What is spin-cross over system? Give examples.	2
2	116.	What is antiferromagnetism?	2
2	117.	Calculate the spin only μ_{eff} values for [K3Fe(CN)6] and [Fe(H2O)6]Cl3.	2
2	118.	Define spin-cross over system? Explain with an example	2
2	119.	arrange the following ligands in order of increasing nephelauxetic effect; H_2O , I^- , F^- , en , CN^-, NH_3	2
2	120.	Δo is always greater than Δt. Comment.	2

3	121.	Briefly discuss the Taube's explanation for lability and inertness of complex	2
3	122.	What are complementary and non-complementary reactions? Give an example for each	2
3	123.	Substitution reaction of $[Cr(CO)_6]$ is very low, consistent with a low spin d^6 complex. But the isoelectronic complex $[V(CO)_5NO]$ is very reactive. Explain	2
3	124.	What are high spin and low spin complexes? Explain with suitable examples	2
3	125.	Why the outer-sphere reactions involving $Ni^{3+/2+}$ are faster than those involving $Co^{3+/2+}$	2
3	126.	Why does a pale pink solution of Co(II) in water turn bright blue on addition of excess hydrochloric acid	2
3	127.	Distinguish between labile and inert complexes.	2
3	128.	The high spin d^4 complex $[Cr(H_2O)_6]^{2+}$ is labile but low spin d^4 complex ion $[Cr(CN)_6]^{4-}$ is inert. Explain	2
3	129.	Define complementary and non-complementary electron transfer reactions	2
3	130.	Arrange the following complexes in the order of increasing lability $[Ca(OH)_6]^{2+}$, $[Sr(OH)_6]^{2+}$, $[Mg(OH)_6]^{2+}$	2
3	131.	Give an example for two electron transfer reactions.	2
3	132.	What are labile complexes? Give an example	2
3	133.	Distinguish between kinetic and thermodynamic stability	2
3	134.	The high spin d^4 complex $[Cr(H_2O)_6]^{2+}$ is labile, but the low spin d^4 complex ion $[Cr(CN)_6]^{4-}$ is inert. Explain	2
3	135.	explain briefly why substitution reactions of $[Ni(CN)_4]^{2-}$ are very rapid.	2
3	136.	Put in order of increasing rate of substitution by water the complexes a) $[Co(NH_3)^6]^{3+}$, $[Rh(NH_3)_3]^{3+}$, $[Ir((NH_3)_6]^{3+}$, $[Mn(OH_2)_6]^{2+}$, $[Ni(OH_2)_6]^{2+}$	2
3	137.	Predict whether the ammonium oxopentachlorochromate (V) with magnetic moment 1.82 would be labile or inert	2
3	138.	Predict whether the potassium hexaiodomanganate (IV) with magnetic moment 3.82 would be labile or inert	2
3	139.	Predict whether the potassium hexacyanoferrate (III) with magnetic moment 2.40 would be labile or inert	2
3	140.	Predict whether the Hexammineiron (II)chloride with magnetic moment 5.45 would be labile or inert	2

3	141.	Explain the mechanistic steps involving SN^1 and SN^2 reactions	2
3	142.	SN^2 path for octahedral complexes is favoured by electron withdrawing ligand. Explain	2
3	143.	Both SN^1 CB and SN^2 mechanism in ligand substitution reaction with a metal complex gives second order kinetics. Justify the statement	2
3	144.	The rate of attack on Co (III) by an entering group is nearly independent on Y. Discuss the mechanism either by associative or dissociative way	2
3	145.	Define chelate effect	2
3	146.	Octahedral complexes of metal centres with high oxidation numbers or of d metals of second and third series are less labile than those of low oxidation number and d metals of the first series of the block. Account for this observation on the basis of dissociative rate determining step.	2
3	147.	Briefly explain the stereochemistry of the substitution in cis complex	2
3	148.	Briefly explain why carbon monoxide has more trans directing ability compared to Cl	2
3	149.	The rate of exchange of anions increases in presence of certain acids in square planar complexes. Explain with suitable example	2
3	150.	Write all the geometrical isomers of the complex $[Pt(NH_3)(Br)(Cl)(py)]$ and how many of these will exhibit optical isomers	2
3	151.	A Pt(II) complex of tetramethyldiethylenetriamine is attacked by Chloride 10^5 less rapidly than the diethylenetriamine analogue. Explain	2
3	152.	How are the charge transfer transition classified. Give an example in each case	2
3	153.	Write the structures of bicapped tetrahedron and monocapped tetrahdedron	2
3	154.	Give an example of pentanuclear cluster and write the structure	2
3	155.	Transition metals in their higher oxidation states do not form M-M bonded compounds, why?	2
3	156.	Write two evidences for M-M bonding	2
3	157.	State Wade-mingo-lauher rules and calculate higher valence electron count for $[Rh_4(CO)_{16}]^{2-}$.	2
3	158.	The cluster $Co_2(C_2H_2)(CO)_6$ has a tetrahedral CoC_2 core. How many electrons does the each cluster contribute to cluster using PSEPT theory.	2

1	159.	Describe any two methods of preparation of metal complexes with an example in each case.	5
1	160.	Describe the preparation of coordination compounds by oxidation reduction and thermal dissociation reaction.	5
1	161.	Discuss the preparation of coordination compounds by addition and substitution methods.	5
1	162.	Explain any two methods adopted for the preparation complex with example.	5
1	163.	Give an account of the preparation of coordination complexes.	5
1	164.	How are the complex formation detected? Explain with examples.	5
1	165.	Discuss the factors affecting the stability of complexes.	5
1	166.	How is overall stability constant of a complex related to stepwise stability constant?	5
1	167.	Explain the factors which affect the stability constant of coordination compounds.	5
1	168.	What is meant by stability of a complex? Explain the factors determining the stability of a complexes.	5
1	169.	Define term overall formation constant? How is step-wise stability constants and over-all stability constant related? Explain why the stability of $Co^{3+} > Co^{2+}$ complex.	5
1	170.	Explain the factors affecting the stability of coordination compounds.	5
1	171.	Write briefly on chelate effect.	5
1	172.	What is Chelate effect? Why it is considered as an entropy effect? Predict which of the following pairs would have higher order of stability. Give reasons.(i) $[Cu(en)_2]^{2+}$ and $[Cu(NH_3)_4]^{2+}$ (ii) [illegible]	5
1	173.	How is stability constant of a complex determined by pH metric method?	5
1	174.	Explain the polarographic method for the determination of the stability constant of a complex.	5
1	175.	How is stability of a complex determined by a spectroscopic method?	5
1	176.	Discuss the any one method of determining the stability constants of complexes.	5
1	177.	Explain the factors affecting the stability of coordination compounds.	5

2	178.	How do the d-orbitals split in a tetraderal and a square planar environment? Explain	5
2	179.	Write the salient features of CFT? Mention its advantages and disadvantages.	5
2	180.	Explain the splitting pattern of d orbital splitting in tetrahedral and octahedral complexes	5
2	181.	Discuss the factors affecting the magnitude of crystal field splitting.	5
2	182.	Define crystal field splitting energy and discuss the splitting of d-orbitals in case of octahedral complexes. Also calculate the CFSE for d5 low spin octahedral complex.	5
2	183.	Define crystal field splitting energy and discuss the splitting of d-orbitals in case of tetrahedral complexes. Also calculate the CFSE for d6 tetrahedral complex.	5
2	184.	Explain the d-orbital splitting takes place in square planar and tetrogonal complexes.	5
2	185.	Describe with evidences for Metal-Ligand orbital overlap.	5
2	186.	Explain the JahnTeller distortion in $[Cu(H2O)_6]^{2+}$.	5
2	187.	What is Jahn-Teller effect? How electronic spectra could be used to detect Jahn-Teller distortion in transition metal complexes?	5
2	188.	Discuss the evidences for metal-ligand covanlency.	5
2	189.	Write a note on metal-ligand covalency.	5
2	190.	Illustrate the factors which explain the evidences of metal-ligand covanlency.	5
2	191.	Explain why Cu2+ does not form a regular octahedral complex whereas Ni2+ does.	5
2	192.	Explain the factors which affect the magnitude of d-orbital splitting values in octahedral complex.	5
2	193.	Set up the MO energy diagram for $[Cr(NH_3)_6]^{3+}$ involving only sigma bonding. Comment on its magnetic property	5
2	194.	Explain with the help of MO theory why Cl^- acts as a weak ligand where as CN^- acts as a strong ligand in octahedral transition metal complexes.	5

2	195.	Depict MO energy level diagram for an octahedral complex involving sigma bonding. Explain its usefulness in accounting for the spectral properties of metal complexes.	5
2	196.	Discuss the effect of π-bonding on the value of Δ in an octahedral complex with ligands capable of forming the π-bond.	5
2	197.	Draw the molecular orbital diagram for ML6 complex involving sigma bonding and explain.	5
2	198.	Set up the MO energy level diagram for an octahedral complex involving both sigma and pi bonding.	5
2	199.	Write a note on selection rules. Explain its relaxation.	5
2	200.	Predict and explain the possible electronic transitions in $[Co(NH_3)_6]^{2+}$	5
2	201.	Write the selection rules for electronic spectra? Why do electronic spectra of $[V(H_2O)_6]^{3+}$ show only two absorption bands against three predicted for the system?	5
2	202.	State and explain the selection rules for elctronic spectra of transition metal complexes.	5
2	203.	Draw and explain the energy level diagram for d2 configuration in tetrahedral and octahedral fields?	5
2	204.	Why does tetrahedral complex of an elemetn give much more intense d-d spectra than its octahedral complex.	5
2	205.	What is electronic spectra? Discuss the electronic spectra of [Ti(H2O)6]3+.	5
2	206.	Draw the absorption spectra of [Ti(H2O)6]3+ and [Cu(H2O)6]2+. Distinguish them on the basis of position of the band and color of the abosrption spectrum	5
2	207.	Draw the Tanabe-Sugano diagram for a d^3 complex	5
2	208.	Sketch and label an orgel diagram for an octahedral d^8 ion. Include the multiplicities in the term symbols.	5
2	209.	How do Tanabe-Sugano diagrams differ from Orgel diagrams? What additional Information is available through T-S diagrams?	5
2	210.	With the Term symbols and draw the Orgel diagram for d^2 octahedral, d^4 tetrahedral complexes.	5

2	211.	Distinguish between Orgel diagrams and Tanabe-Sugano diagrams.	5
2	212.	Draw and explain the Orgel diagram for d^2, d^7 octahedral and d^8, d^3 tetrahedral configuration of metal ion and explain the possible electronic transitions.	5
2	213.	With the term symbols draw the Orgel diagram for d^1 and d^6 tetrahedral and octahedral complexes.	5
2	214.	What are Tanabe-Sugano diagrams? In what way they are superior to Orgel diagrams?	5
2	215.	What are Tanabe-Sugano diagrams? Write Tanabe-Sugano diagram for $[Ni(H_2O)_6]^{2+}$.	5
2	216.	Draw the Orgel diagrams for d2 octahedral and d4 tetrahedral complex and explain the possible electronic transitions.	5
2	217.	Set up the MO diagram for [Cr(NH3)6]3+ involving only sigma bonding. Comment on its magnetic property.	5
2	218.	Sketch and label an orgel diagram for an octahedral d8 ion. Include the multiplicities in the term symbols.	5
2	219.	Draw the combined Orgel diagram for d1, d4, d6, d9 complexes.	5
2	220.	What are charge transfer transitions? Explain.	5
2	221.	Give an account of charge transfer spectra.	5
2	222.	Write short notes on Nephelauxetic effect	5
2	223.	How are charge transfer tranistions classified? Give example in each case.	5
2	224.	Write a note on charge transfer spectra.	5
2	225.	explain the terms nephelauxetic effect and nephelauxetic ration. What is the significance of nephelauxetic effect?	5
2	226.	What is magnetic susceptibility? Explain how this is measured by Gouy's method.	5
2	227.	What is meant by magnetic susceptibility of a substance? Describe the use of Gouy method for its determination.	5
2	228.	Discuss the effect of temperature on the magnetic susceptibility of transition metal complexes	5

2	229.	Write a short notes on a) Paramagnetism b) Diamagnetism c) Ferro and Ferromagnetism	5
2	230.	Explain temperature independent paramagnetism with suitable examples.	5
2	231.	With examples distinguish among ferri and ferro magnetic substances.	5
2	232.	Write short notes on ferromagnetism and antiferromagnetism.	5
2	233.	Write a note on ferromanetism, Curie temperature, L-S coupling.	5
2	234.	State and explain the Jahn-Teller effect? What are its consequenses?	5
3	235.	Explain the Taube mechanism of inner sphere electron transfer reaction in metal complexes	5
3	236.	Discuss valence bond explanation and taube's theory for lability and inertness of a complex	5
3	237.	Distinguish between SN_2 and SN_1 CB mechanism for the base hydrolysis of Co(III) complex	5
3	238.	Discuss in detail about the mechanism involved in base hydrolysis of Cobalt (III) ammine complexes	5
3	239.	Distingish between SN_2 and SN_1 CB mechanism for the base hydrolysis of Cobalt (III) complex	5
3	240.	Explain the mechanism of substitution reactions for octahedral complexes	5
3	241.	Give the differences between SN^1 and SN^2 mechanism for substitution reaction in octahedral complexes	5
3	242.	Predict the product and suggest mechanism for the following conversion by SN^1 CB ; $[Co(NH_3)X]^{2+} + OH^-$ ⟶	5
3	243.	Discuss the factors affecting the rate of reaction in octahedral complexes	5
3	244.	Discuss in detail the various factors which affect the stability of metal complexes	5
3	245.	Explain the chelate effect for nucleophillic substitution reaction	5
3	246.	Explain the mechanism of anation reaction in octahedral complexes	5
3	247.	Explain the preparation of cis and trans form of $[Pt(C_2H_4)(NH_3)Cl_2]^0$	5
3	248.	Explain the preparation of cis and trans form of $[Pt(NH_3)_2Cl_2]^0$	5
3	249.	What is trans effect? How is it explain by polarization theory	5

3	250.	Discuss the trans effect in ligand substitution in square complexes	5
3	251.	The complex $[PtCl_2(NH_3)(NO_2)]$ can be prepared specifically as both cis and trans isomers by making use of trans effect. Given that the trans effect for the three ligands here are nitrite > chloride > ammonia, suggest an approach to synthesise each isomer	5
3	252.	Discuss the mechanism of ligand substitution reaction of square planar complexes	5
3	253.	Explain the trans effect in the ligand substitution in square planar complex	5
3	254.	What is trans effect? Suggest the reaction pathways for preparing three isomers of $[Pt(py)NH_3BrCl]$ from $[PtCl_4]^{2-}$	5
3	255.	What is trans effect? What would be the sequence of substitution required to get $[Pt(Cl)(Br)(NH_3)(Py)]$ from $[PtCl_4]^{2-}$.	5
3	256.	How are cis and trans $[PtCl_2(NH_3)(NO_2)]^-$ synthesized from $[PtCl_4]^{2-}$	5
3	257.	Explain the polarization theory for trans substitution effect	5
3	258.	Explain π-bonding and polarization theory of trans substitution reaction	5
3	259.	Explain the π bonding theory of substitution in platinum complexes	5
3	260.	Explain polarization theory and π-bonding theory for trans substitution in platinum complexes	5
3	261.	How does each of the following modification affect the rate of square planar complex substitution reaction? a) Changing a trans ligand from H to Cl b) Changing the leaving group from Cl to I c) adding a bulky substituent to a cis ligand d) increasing the positive charge on the complex	5
3	262.	Predict the product of the following reactions a) $[Pt(PR_3)]^{2+} + 2\ Cl^-$; b) $[PtCl_4]^{2-} + 2PR_3$; cis-$[Pt(NH_3)_2(py)_2]^{2+} + 2Cl^-$	5
3	263.	When the iosmers of $[Pt(NH3)_2Cl_2]$ react with thiourea (tu), one product is $[Pt(tu)_4]^{2+}$ and the other is $[Pt(NH_3)(tu)_2]^{2+}$. Identify the initial isomers and explain the results	5
3	264.	Predict the product (equimolar mixture), include each step of the reaction process a) $[Pt(CO)Cl_3]^- + NH_3$; b) $[Pt(NH_3)\ Br_3]^- + NH_3$; c) $[(C_2H_4)\ Pt\ Cl_3]^- + NH_3$	5
3	265.	Design a sequence of reactions, beginning with $[PtCl_4]^{2-}$, that will result in platinum (II) complexes with four ligands , py, NH_3, NO_2^-, CH_3NH_2. assuming that the transeffect of methylamine and ammonia are similar	5
3	266.	Design two step synthesis of cis and trans $[PtCl_2(NO_2)(NH_3)]^-$ starting from $[PtCl_4]^{2-}$	5

3	267.	Explain outer sphere mechanism with suitable examples	5
3	268.	Explain with examples of outer sphere mechanism of substitution reaction in octahedral complexes	5
3	269.	Explain outersphere electron transfer reaction on the basis of marcus theory	5
3	270.	The electron transfer between $[Co(NH_3)_5Cl]^{2+}$ and $[Cr(H_2O)_6]^{2+}$ leads to quantitative transfer of chloride ion from cobalt to chromium. Explain the mechanism involved in its conversion. How this conversion is confirmed experimentally.	5
3	271.	Write the inner and outer sphere pathways for reduction of azidopentaaminecobalt (III) ion with $V^{2+}_{(aq)}$. What experimental data might be used to distinguish between the two pathways?	5
3	272.	Discuss the electron transfer by inner sphere mechanism for the following reaction; Reaction between $[Co(NH_3)_5Cl]^{2+}$ with $[Cr(H_2O)_6]^{2+}$ in which chloride ion exchanges between Cobalt and chromium complex	5
3	273.	Explain the detailed mechanism in the interconversion of ferrocyanide/ ferricyanide by outersphere mechanism	5
3	274.	Explain the detailed mechanism in the interconversion of $[Co(NH_3)_6]^{2+}$ /$[Co(NH_3)_6]^{3+}$ by outersphere mechanism	5
3	275.	State wade-mingo's-Lauher rule. Illustrate its use by taking an example of metal cluster	5
3	276.	Explain the factors favoring the metal-metal bonding in metal clusters	5
3	277.	State wades-mingos-lauher rules and calculate higher valence electron count for $[Re_4(CO)_{16}]^{2-}$	5
3	278.	Rationalize why $Os_3(CO)_{12}$ has a triangular Os_3 core but $Os_3(CO)_{12}Br_2$, the Os atoms are in linear arrangement	5
3	279.	Explain the factors which favour M-M bond formation. What are the evidences which suggest the existence of M-M in a complex.	5
3	280.	Explain the factors favoring M-M bond formation	5
3	281.	What are wade-Mingos-Lauher rules? Calculate higher valence electron count for $[Re_4(CO)_{16}]^{2-}$ and $[Ru_6C(CO)_{17}]$	5
3	282.	In $Os_4(CO)_{16}$, the osmium atoms are arranged in a square planar, but in $Os_4(CO)_{14}$ form tetrahedral cluster. Rationalize this observation. Depict the structure of each cluster.	5

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Question Paper Pattern- Model Question Paper

Q.P Code: 14MSCBH120

St. Philomena's College (Autonomous) Mysore

II Semester M.Sc. Chemistry Supplementary Examination May - 2018

Subject: INORGANIC CHEMISTRY

Title: Co-ordination Chemistry

Time: 3 Hours **Max Marks: 70**

Instruction to the Candidates: Question paper has 2 parts answer both the parts. Provide clear and concise answer to the questions. Use chemical equations / drawings to support your answer where ever appropriate.

PART –A

1. **Answer the following questions:** **10x2=20**

a. Account for the general trend, the step-wise stability constants gradually decreases.

b. What is Irwing-William series? Explain with example.

c. What are highspin and low –spin complexes Give examples.

d. Briefly discuss the Taube's explanation for liability and inertness of complex.

e. What are complementary and non complementary reactions Give an example for each?

f. Define outer sphere mechanism with an example.

g. How do d-orbitals split in square planar and tetragonal geometry? Write the increasing order of energies of the d-orbitals ?

h. For $[Cr(H_20)_6]^{2+}$ ion Ep=23500 cm^{-1} and $\Delta 0 = 13900\ cm^{-1}$ will it be a high spin or low spin complex?

i. Rationalize the fact that the stability increases in the order $[Co(NH_3)_6]^{3+} < [Co(en)_3]^{3+} < [Co(dien)_2]^{3+}$

j. Write the structures of bicapped tetrahedron and monocapped octahedron.

PART –B

Answer any five of the following: **5x10=50**

2. a. Explain the preparation of cis and trans form of $[Pt(C_2H_4)(NH_3)Cl_2]^0$ (5+5)

b. How is the overall stability constant of a complex related to its step-wise stability constant? 5+5

PTO

3. a. Discuss the electron transfer by inner sphere mechanism for the following reaction (5+5)

$[Co(NH_3)_5Cl]^{2+} + [Cr(H_20)_6]^{2+} + 5H_30^+ \longrightarrow [Co(H_2O)_6]^{2+} + [Cr(H_2O)_5Cl]^{2+} + 5NH_4^+$

b. Discuss the factors affecting the rate of reaction in octahedral complexes.

4. a. What are the different methods adopted for the preparation of coordination complexes? Explain with examples. (5+5)

b. Explain the polarization theory for trans substitution effect.

5. a. Explain the factors affecting the stability of coordination compounds. (5+5)

b. Discuss the determination of stability constant of a complex by pH metric method.

6. a. With the team symbols draw the Orgel diagram for d^1, d^6 tetrahedral and octahedral complexes

b. State and explain the Jahn-Tellers theorem? What are its consequences? (5+5)

7. a. Explain the mechanism for the following conversion by conjugate base mechanism $[Co(NH_3)_5Cl]^{2+} \longrightarrow [Co(NH_3)_5OH]^{2+}$ (5+5)

b. Illustrate the factors which explain the evidences of metal-ligand convalency 5+5

8. a. Draw the M.O diagram for $[Cr(NH_3)_6]^{3+}$.Comment on the magnetic properties

b. Draw and the Tanabe-Suguno diagram for the d^3 System (5+5)

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